

"I Feel More Worried About My Privacy" Public Perceptions of Biometric Traces in Everyday Interactions

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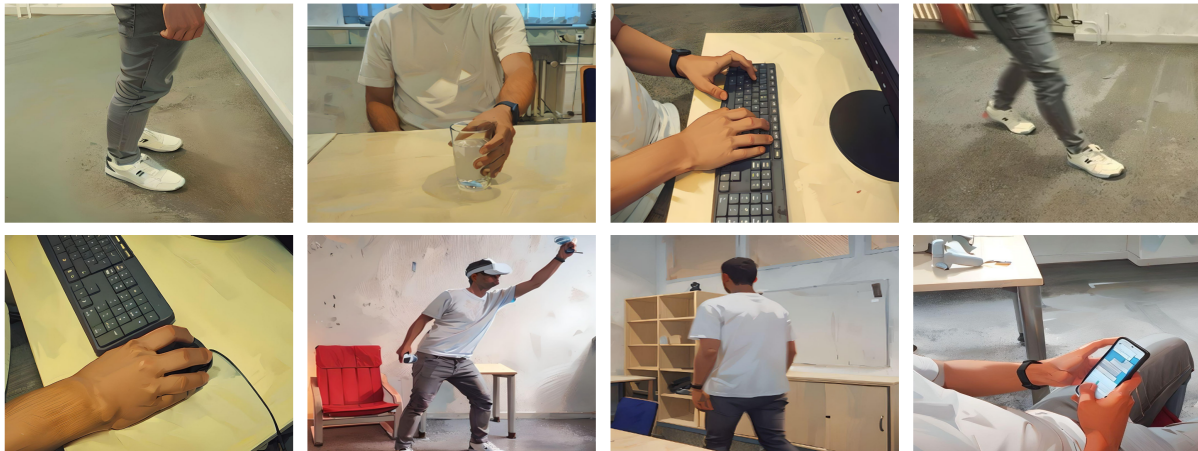


Figure 1: Overview of the eight biometric modalities examined in this study. Top (L→R): thermal traces from standing; fingerprints on glass; typing (keystroke dynamics); walking (gait). Bottom (L→R): mouse movements; interactions in VR; face captured by a shelf-mounted camera; touch gestures on a smartphone.

Abstract

People leave behind biometric traces through everyday interactions, often without fully understanding their implications. While biometric technologies increasingly rely on subtle behavioral and interaction-based signals, little is known about how people perceive these traces in daily life. We present findings from an exploratory online study (N=120) that used short, scenario-based videos to illustrate situations in which biometric traces may be inadvertently

exposed, including fingerprints, gait, thermal residues, and interaction patterns in virtual environments. We examine which traces people recognize, how concerned they are about potential misuse, and how brief exposure to such scenarios shapes privacy perception. Results show that awareness and concern frequently diverge. Participants were familiar with visible, well-known biometrics, yet less aware of emerging or interaction-borne traces. Importantly, exposure to the scenarios prompted several participants to reconsider their privacy assumptions.



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CCS Concepts

• **Human-centered computing** → *Empirical studies in ubiquitous and mobile computing*; • **Security and privacy** → *Social aspects of security and privacy*.

Keywords

biometric traces, behavioral biometrics, privacy risks, user awareness, usable privacy

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1 Introduction

Touching a glass, walking across a room, typing on a keyboard, or playing a virtual reality game are routine actions that rarely prompt privacy concerns. Yet each of these interactions can leave behind biometric traces, physiological or behavioral characteristics that can be used to identify an individual, often in the form of residual signals unintentionally produced during the interaction [13]. Unlike passwords or explicit authentication mechanisms, such traces are not deliberately provided but emerge implicitly, without clear moments of awareness or consent.

As biometric sensing becomes increasingly embedded in everyday technologies, identification no longer relies solely on deliberate actions such as unlocking a device with a fingerprint or face [18, 28]. Instead, identity can be inferred from behavioral patterns, interaction styles, and residual signals produced through ordinary use [34]. This shift raises a fundamental question: *what do people believe they reveal about themselves through everyday interactions, and when do they perceive this as a privacy risk?*

While biometric systems increasingly support implicit and continuous identification, users are often expected to make informed privacy decisions about processes that remain largely invisible. This expectation is misaligned with public understanding, as many biometric traces are unnoticed or underestimated until they are explicitly surfaced. Such gaps challenge meaningful consent, trust, and privacy, particularly in shared or public settings where biometric traces may be exposed to others.

To examine how non-expert users, i.e., individuals without professional expertise in biometrics, security, or privacy technologies, interpret biometric traces in everyday contexts, we adopt a scenario-based approach using short videos depicting routine interactions. By prompting participants to reflect on these scenarios before and after highlighting specific biometric traits, we investigate both what is recognized as identifying and how brief, situated exposure shapes privacy perception. Our findings provide empirical insights into the awareness–concern gap surrounding biometric traces and highlight scenario-based exposure as a lightweight means to surface otherwise hidden privacy risks.

2 Related Work

Biometric identification relies on physiological and behavioral traits to distinguish individuals, offering an alternative to knowledge- or possession-based authentication mechanisms such as passwords or tokens [12, 14, 35]. To qualify as a biometric, a trait must be distinctive yet universal, sufficiently stable over time, and measurable in a quantitative manner [12]. While biometrics are often

valued for their convenience and robustness, their immutability poses significant privacy risks: once biometric data are leaked or misused, they cannot be easily revoked or replaced. Physiological biometrics such as fingerprints and facial recognition are widely studied and integrated into consumer devices [10, 15, 22, 28, 38, 41]. Other physiological traits, including iris [2, 27], palm [4, 26], and footprints [17, 25], have also been explored but are less pervasive. In parallel, research on behavioral biometrics has grown substantially, focusing on distinctive patterns in gait [5, 21], keystroke dynamics [8, 39], mouse movements [16, 42], and touch interaction on mobile devices [6, 40]. More recently, biometric identification has expanded into emerging contexts such as virtual reality, where body motion and task-based interaction patterns can serve as identifying signals [1, 19, 32], as well as thermal imaging, where residual heat traces enable identification under certain conditions [36].

Despite this growing technical literature, comparatively less attention has been paid to how non-expert users perceive biometric identification and its privacy implications. Prior work has examined privacy awareness and concern across a range of digital environments, particularly in XR and VR contexts [29, 31]. Survey and interview studies indicate that users often have limited awareness of the types of data collected about them, which constrains their ability to adopt privacy-protective strategies [9, 20]. Interactive interventions, such as VR privacy tutorials or permission-management interfaces, have shown potential to increase awareness and satisfaction by making data practices more explicit [20, 37]. Large-scale surveys of smartphone users also reveal low levels of security and privacy awareness among the general public [3]. Findings suggest that awareness is frequently incomplete and uneven, even for widely deployed technologies.

In parallel, research on interaction traces has explored how traces of user activity, such as walking trajectories, movement patterns, or interaction histories, can support social awareness and coordination in shared or public environments [7, 11, 24, 30, 33]. While such traces are often designed to enhance collaboration or presence, they can also function as implicit identifiers. However, existing work rarely foregrounds the privacy risks associated with interaction traces being observed, recorded, or repurposed for identification. Our work builds on these strands by focusing explicitly on how people perceive the biometric traces they leave behind during everyday interactions, and how awareness and concern emerge when such traces are made visible.

In summary, prior work has examined both the prevalence of biometric-based identification and persistent gaps in privacy awareness; however, little is known about how users perceive biometric traces that emerge from routine, everyday interactions. Addressing this gap motivates our exploratory investigation.

3 Online Survey: Users' Perceptions of Biometric Traces in Everyday Interactions

To understand how non-expert users perceive biometric traces in everyday interactions, we conducted an exploratory, scenario-based online survey focusing on users' interpretations and concerns rather than biometric system performance.

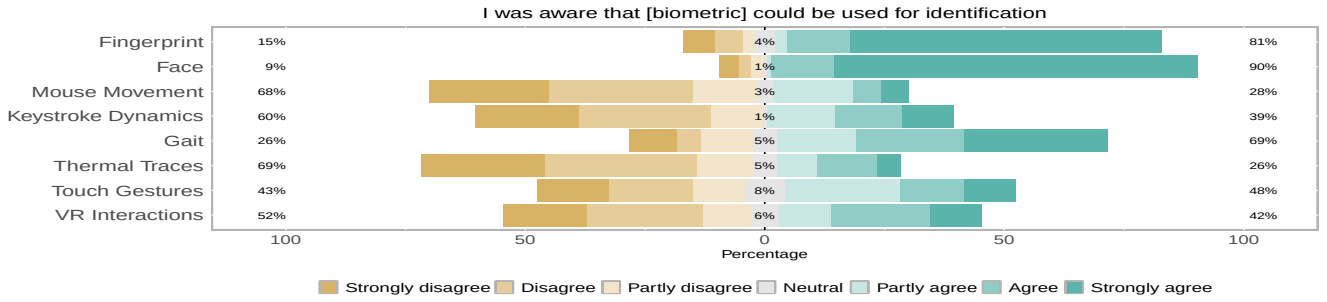


Figure 2: Participant awareness of eight biometric traits as identifiers.

Survey Design and Procedure. The survey employed a staged, scenario-based design to elicit perceptions grounded in everyday experience. Participants were shown four short videos (15–48 seconds) depicting routine home and office interactions in which biometric traces may be inadvertently produced (e.g., touching objects, walking, typing, or interacting in virtual environments)¹. The videos were developed by the authors to illustrate eight target biometric modalities within realistic office-based interactions. Each modality was embedded in a concrete action to demonstrate how the trace could be generated, without relying on technical explanations or prior expertise. The authors iteratively reviewed the videos to ensure that the intended biometric traces were clearly depicted and comparably framed across scenarios. Across the scenarios, we covered eight biometric modalities spanning a range of familiarity: two physiological traits (*face*, *fingerprints*), four behavioral traits (*gait*, *keystroke dynamics*, *mouse movements*, *touch gestures*), and two emerging identifiers (*thermal traces of feet* [36] and *VR interaction patterns* [19]). The modalities were not explicitly named in the videos; instead, participants inferred what could be used for identification.

After providing informed consent, participants first completed a demographics questionnaire. They then viewed the four videos in randomized order, one after another, and answered open-ended questions describing what they observed and which characteristics they believed could be used for identification, capturing unprompted perceptions. After all videos had been shown, participants were presented with still images highlighting the biometric traits of interest and rated their awareness of each modality using 7-point Likert scales. In the final phase, participants received a general definition of biometrics, described potential misuse or attack scenarios for each modality in free text, and rated their level of concern. The survey concluded with optional comments and a debriefing. An attention check was included, and navigation was one-way only to prevent revision of earlier responses.

Measures. We collected both quantitative and qualitative measures to assess participants’ *awareness*, *concern*, and *understanding of privacy risks* related to each biometric modality. For each video, participants first answered an open-ended question: “What specific actions or characteristics are unique enough to identify a person? (What in the video reveals who someone is?)”. This free-response

item captured spontaneous *awareness* or recognition of potentially identifying traits without priming by technical terms such as “biometrics.” Afterward, participants rated their agreement with the statement “I was aware that this characteristic could be used for identification” on a 7-point Likert scale (1=strongly disagree, 7=strongly agree). In a later section, we provided a general definition of biometrics following Jain et al. [12] and informed participants that biometric data may be targeted by adversaries. For each modality, they described in free text how such data could be compromised (e.g., spoofing or mimicry). This measure probed participants’ familiarity with *threat models*. Participants rated their level of *concern* for each biometric trait using the prompt: “I am concerned about the potential risks of my [trait name] being recorded and analyzed.” Ratings were again given on a 7-point Likert scale (1=very unconcerned, 7=very concerned). At the end of the survey, participants could provide open-ended feedback, allowing qualitative insights into reflections, misunderstandings, or shifts in perception.

Participants. We recruited 120 participants via the Prolific crowdsourcing platform in the summer of 2024. Eligibility criteria required participants to be at least 18 years old, fluent in English, and not to have participated in related pilot studies. No professional expertise in biometrics, security, or privacy was required, as the study targeted non-expert members of the general public. The sample included 65 participants identifying as female, 54 as male, and one who did not specify gender, with ages ranging from 18 to 72 years ($M = 32.55$, $SD = 11.58$). Participants reported diverse employment statuses, including full-time (41), part-time (27), unemployed (23), and other arrangements (29; e.g., student, self-employed, retired). They represented a broad range of countries, with the largest groups residing in Canada ($n=22$), the United Kingdom ($n=17$), Chile ($n=15$), and Mexico ($n=15$), alongside participants from the United States, Spain, Germany, Portugal, Italy, Poland, and several other countries. As with many crowdsourced studies, the sample may not fully represent the general population, and findings should be interpreted in light of this limitation. The median survey completion time was 23 minutes, and participants were compensated at Prolific’s recommended minimum rate. The study was conducted in accordance with institutional ethics guidelines, and informed consent was obtained from all participants.

¹Videos and corresponding images are available here: https://osf.io/f49us/files/osfstorage?view_only=916775fcd684275938c6617854aa108

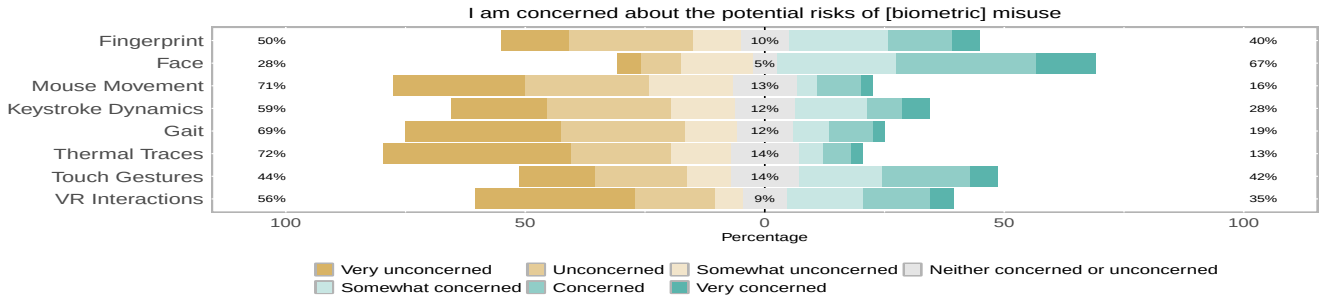


Figure 3: Participant concern about potential misuse of biometric traits.

Data Analysis. We analyzed the data using a combination of qualitative and quantitative methods to capture both the richness of participant reflections and measurable patterns in awareness and concern. Open-ended responses from the awareness, attack knowledge, and optional comment questions were analyzed using inductive thematic analysis [23]. Three researchers independently open-coded a randomly selected subset of responses ($n=20$ per researcher). We then held a review session to discuss code interpretations, resolve discrepancies, and consolidate the coding tree. The agreed-upon structure was applied to the full dataset, with categories refined into broader themes that characterize participant perceptions and concerns. This approach allowed unexpected privacy-relevant insights to emerge beyond the predefined biometric modalities. All qualitative coding was conducted using MAXQDA². On the other hand, Likert-scale items for awareness and concern were treated as ordinal data. We therefore applied non-parametric statistical tests, using Wilcoxon signed-rank tests for within-modality comparisons and Kruskal–Wallis tests for between-modality differences. Where applicable, post-hoc pairwise comparisons were Bonferroni-corrected. All statistical analyses were performed in R (version 2024.04.2+764).

4 Results

We report findings on participants’ awareness of different biometric traces, their concern about potential misuse, and how these perceptions relate. We complement the quantitative results with qualitative reflections that capture shifts in privacy perception following scenario-based exposure.

Awareness of Biometric Traces. Participants’ awareness of biometric traces varied strongly across modalities and was primarily shaped by visibility and familiarity rather than by whether a trace arises implicitly through interaction. Visible and widely publicized identifiers were readily recognized as biometric, whereas interaction-borne and emerging traces were frequently overlooked.

This pattern is reflected in participants’ awareness ratings after highlighted exposure (Figure 2). Conventional physiological identifiers such as *face* ($M = 6.33$, $SD = 1.54$) and *fingerprints* ($M = 5.89$, $SD = 1.92$) received the highest awareness scores, followed by *walking patterns* ($M = 5.02$, $SD = 1.99$). Awareness was more mixed for *touch gestures* ($M = 3.90$, $SD = 1.98$) and *VR interactions*

($M = 3.73$, $SD = 2.13$), indicating heterogeneous familiarity. In contrast, awareness was lowest for less visible and interaction-borne traces, including *keystroke dynamics* ($M = 3.41$, $SD = 2.12$), *mouse movements* ($M = 2.95$, $SD = 1.86$), and *thermal traces* ($M = 2.95$, $SD = 1.92$). A Friedman test confirmed significant differences in awareness across modalities ($\chi^2(7) = 319.76$, $p < .001$). Post-hoc Wilcoxon tests showed that face recognition was rated significantly higher than all other modalities. Fingerprints and walking patterns formed a second group of familiar identifiers, while no significant differences emerged among keystroke dynamics, thermal traces, touch gestures, and VR interactions. Taken together, these results suggest that *public awareness of biometric traces is uneven and closely tied to how visible and well-established a modality is, rather than to whether it can function as a biometric identifier in practice*.

Concern About Potential Misuse. Participants’ concern about biometric misuse followed a different pattern than awareness, being shaped more by perceived personal salience and direct identifiability than by familiarity. This pattern is reflected in the concern ratings shown in Figure 3. *Face* recognition elicited the highest level of concern ($M = 4.72$, $SD = 1.71$), followed by *touch gestures* ($M = 3.78$, $SD = 1.92$) and *fingerprints* ($M = 3.63$, $SD = 1.87$). Lower concern was reported for *keystroke dynamics* ($M = 3.21$, $SD = 1.84$) and *VR interactions* ($M = 3.21$, $SD = 2.07$), with the lowest concern associated with *mouse movements* ($M = 2.79$, $SD = 1.69$), *gait* ($M = 2.74$, $SD = 1.77$), and *thermal traces* ($M = 2.53$, $SD = 1.68$). A Friedman test confirmed significant differences in concern across biometric modalities ($\chi^2(7) = 201.84$, $p < .001$). Post-hoc Wilcoxon tests with Holm–Bonferroni corrections showed that concern for *face* was significantly higher than for all other modalities. *Touch gestures* and *fingerprints* elicited significantly higher concern than *mouse movements*, *gait*, and *thermal traces*, while concern for *thermal traces* was significantly lower than for *VR interactions*. Overall, these findings indicate that *privacy concern is not uniformly distributed across biometric traces and does not simply follow awareness*. Instead, concern appears to be shaped by how directly a trace is associated with personal identity and by how intrusive or consequential its capture is perceived to be.

Awareness vs. Concern. Taken together, the results reveal a clear disconnect between awareness and concern across biometric modalities. Knowing that a trait can function as a biometric identifier did not reliably translate into heightened concern about its potential misuse. This pattern is reflected in weak associations between

²<https://www.maxqda.com/>

awareness and concern ratings across all modalities. Spearman's rank correlations were small in magnitude (all $|\rho| \leq .29$), indicating little to no relationship between being aware of a biometric trace and perceiving it as risky (e.g., *fingerprints* $\rho \approx -0.07$; *face* $\rho \approx 0.01$; *touch gestures* $\rho \approx 0.29$). These findings suggest *that awareness alone is insufficient to predict privacy sensitivity*. At the same time, participants' qualitative feedback points to a more nuanced dynamic. Several participants described the survey as an "eye-opener," noting that exposure to the scenarios prompted them to reconsider everyday interactions as potential sources of biometric risk. Comments such as "I feel more worried about my privacy" and "This study touched on something I had never really considered" illustrate how *brief, situated exposure can surface latent concerns that are not captured by quantitative correlations alone*. Together, these results indicate that while awareness and concern are weakly coupled at a population level, reflective exposure to concrete scenarios can challenge existing assumptions and reshape how users perceive biometric risks.

Knowledge of Potential Attacks. To contextualize awareness and concern, we analyzed participants' descriptions of potential attack vectors. Responses revealed limited understanding of how biometric traces could be misused. Familiar modalities such as *face* and *fingerprints* were associated with capture and spoofing attacks, whereas less visible traces (e.g., *thermal*, *VR*, *mouse*) elicited uncertainty and speculative responses. *This pattern mirrors the observed awareness–concern gap and indicates that emerging, interaction-borne traces remain weakly represented in users' threat perceptions.*

5 Conclusion

This work examined how non-expert users perceive biometric identifiers across both established modalities (face, fingerprints, gait) and emerging, interaction-borne traces (thermal residues, VR interactions, keystroke and mouse dynamics, touch gestures). Guided by our research questions, we investigated (i) whether people recognize the biometric traces they leave, (ii) how concerned they are about their misuse, and (iii) how awareness and concern relate, complemented by participants' knowledge of potential attacks. Our findings show that participants are highly aware of visible, widely publicized identifiers, but far less familiar with novel, less-visible traces. Awareness did not consistently translate into concern, underscoring a gap between recognition and risk sensitivity. Users most often envisioned capture and spoofing attacks, while malware and emerging modalities were more difficult to imagine, revealing fragmented and incomplete mental models. Importantly, qualitative feedback indicated that scenario-based exposure can shift perceptions, suggesting a promising pathway for awareness interventions. These results highlight critical vulnerabilities and opportunities for privacy-enhancing design. Protecting users against biometric leakage means not only strengthening technical safeguards, but also making hidden traces and risks understandable in ways that align with everyday interactions.

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used generative AI-based tools to revise the text, improve flow, and correct any editorial issues.

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A Survey Questionnaire

The online survey consisted of four main parts and combined open-ended and Likert-scale items. The complete questionnaire is available upon request.

A.1 Video-Based Perception Tasks

Participants watched four short videos depicting everyday interactions (e.g., touching objects, walking, interacting with devices, and using virtual reality). After each video, participants answered two open-ended questions: (1) a brief description of what they observed, and (2) which actions or characteristics in the video could be used to identify a person. These questions captured unprompted perceptions of potentially identifying traits.

A.2 Awareness of Biometric Traces

After all videos had been shown, participants were presented with still images highlighting specific biometric traits and informed that these traits *could* be used for identification. For each modality (face, fingerprints, gait, mouse movements, keystroke dynamics, touch gestures, virtual reality interactions, and thermal traces), participants rated their awareness using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). An attention check item was included.

A.3 Concern About Potential Misuse

Participants rated their level of concern for each biometric modality using a 7-point Likert scale (1 = very unconcerned, 7 = very concerned), responding to prompts about the potential risks of each trait being recorded and analyzed.

A.4 Knowledge of Potential Attacks

Participants answered open-ended questions describing how each biometric modality could be misused or attacked (e.g., spoofing, behavioral imitation, or data theft).

A.5 Optional Feedback

At the end of the survey, participants could provide additional comments or reflections in a free-text field.